

MATHEMATICS – YEAR 7

Achievement Standard Guide



About this Achievement Standard Guide

This Achievement Standard Guide describes the expected knowledge, skills and understanding students should demonstrate for Mathematics in Year 7.

The Guides were developed using the achievement standards of Version 9.0 of the Australian Curriculum. The achievement standards assist teachers to monitor student learning and to make judgements about student progress and achievement.

The Guides set out the aspects of the achievement standard and describe indicators at five levels of attainment. They identify possible observable characteristics that could demonstrate the level of the student's understanding.

The Achievement Standard Guides were developed to:

- support teacher judgement when assessing student learning
- support planning for learning in all aspects of the achievement standard
- support the development of clear and specific learning intentions and success criteria.

The Achievement Standard Guides can be used to:

- ensure consistency of assessment
- support moderation of student work at year, school, network and state level
- support unit and lesson planning
- ensure teachers have sufficient evidence of learning to make defensible on-balance judgements about the quality of students' work against the achievement standards

An effective assessment program will:

- Guide and support targeted teaching and learning
- Ensure students have opportunities to demonstrate depth and breadth of their learning in all aspects of the achievement standard
- Provide regular feedback to student about how they can improve in their learning
- Clarify future teaching and learning needs

The following information supports teachers in these aspects of quality assessment practice.

Mathematics Year 7 Achievement Standard

Strand	By the end of Year 7, students:
<u>Number</u>	• represent natural numbers in expanded form and as products of prime factors, using exponent notation • solve problems involving squares of numbers and square roots of perfect square numbers • solve problems involving addition and subtraction of integers • use all 4 operations in calculations involving positive fractions and decimals, choosing efficient calculation strategies • choose between equivalent representations of rational numbers and percentages to assist in calculations • use mathematical modelling to solve practical problems involving rational numbers, percentages and ratios, in financial and other applied contexts, justifying choices of representation
<u>Algebra</u>	• use algebraic expressions to represent situations, describe the relationships between variables from authentic data and substitute values into formulas to determine unknown values a) use algebraic expressions to represent situations b) describe the relationships between variables from authentic data c) substitute values into formulas to determine unknown values • solve linear equations with natural number solutions • create tables of values related to algebraic expressions and formulas, and describe the effect of variation.
<u>Measurement</u>	• apply knowledge of angle relationships and the sum of angles in a triangle to solve problems, giving reasons. • use formulas for the areas of triangles and parallelograms and the volumes of rectangular and triangular prisms to solve problems • describe the relationships between the radius, diameter and circumference of a circle
<u>Space</u>	• classify polygons according to their features and create an algorithm designed to sort and classify shapes • represent objects two-dimensionally in different ways, describing the usefulness of these representations • use coordinates to describe transformations of points in the plane
<u>Statistics</u>	• plan and conduct statistical investigations involving discrete and continuous numerical data, using appropriate displays • interpret data in terms of the shape of distribution and summary statistics, identifying possible outliers • decide which measure of central tendency is most suitable and explain their reasoning
<u>Probability</u>	• list sample spaces for single step experiments, assign probabilities to outcomes and predict relative frequencies for related events • conduct repeated single-step chance experiments and run simulations using digital tools, giving reasons for differences between predicted and observed results.

How the Achievement Standard Guide is organised

The relevant achievement standard statement in italics.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	Achievement standard statement with bolded variations For example, students can: • ...	Achievement standard statement For example, students can: • ...	Achievement standard statement with bolded variations For example, students can: • ...	

Content strand
and sub-strand

'Can do' statements associated with each attachment level, including possible observable characteristics as examples of what students can do.

Blank spaces or merged cells indicates a specific skill that can either be achieved or not

 Merged cell indicates that attainment is possible beyond 'At Standard' but not readily discernible between 'Above' or 'Well Above'.

Number

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Number	With support, students represent natural numbers in expanded form and as products of some prime factors, possibly using some exponent notation. For example, students can: Attempt a partial prime factorisation or expanded form. Eg $360 = 300 + 60$ $= 3 \times 100 + 6 \times 10$ $= 3 \times 10 \times 10 + 6 \times 10$ $= 3 \times 5 \times 2 \times 5 \times 2 + 6 \times 5 \times 2$	Students represent natural numbers in expanded form and as products of prime factors, using exponent notation. For example, students can: - Represent 360 in expanded form: $360 = 300 + 60$ $= 3 \times 10^2 + 6 \times 10^1$ Prime factorisation. $360 = 2 \times 180$ $= 2 \times 2 \times 90$ $= 2 \times 2 \times 2 \times 45$ $= 2 \times 2 \times 2 \times 9 \times 5$ $= 2 \times 2 \times 2 \times 3 \times 3 \times 5$ $= 2^3 \times 3^2 \times 5^1$	Students systematically and efficiently represent natural numbers in expanded form and as products of prime factors, using exponent notation to find sets of factor pairs. For example, students can: - Use prime factorisation to find other factor pairs. Eg correctly factorise 360: $= 36 \times 10$ $= 6 \times 6 \times 5 \times 2$ $= 3 \times 2 \times 3 \times 2 \times 5 \times 2$ $= 2^3 \times 3^2 \times 5^1$ - Use this to find factor pairs such as 15×24, 18×20 etc.	Students systematically and efficiently represent natural numbers in expanded form and as products of prime factors, using exponent notation to solve problems related to factor pairs. For example, students can: Use prime factorisation to find other factor pairs and factors that are multiple of a particular factor. Eg what are some numbers that are not only multiples of 3 but also factors of 720?
	Students solve and use symbols for one-step problems involving squares of numbers and square roots of perfect square numbers up to 100. For example, students can: Find perfect squares up to 10^2 and square roots of perfect squares up to 100 eg: a) $9^2 = 81$ b) $\sqrt{64} = 8$	Students solve problems involving squares of numbers and square roots of perfect square numbers. For example, students can: - Find perfect squares up to 20^2 without using a calculator, and can justify solutions through two different representations (including diagrams and physical materials). - Approximate the square roots of non-perfect squares, up to 100. Eg 40 lies between two perfect squares, what are they, and between which 2 consecutive natural numbers must $\sqrt{40}$ be? Explain how you know this is true.	Students flexibly solve problems involving squares of numbers and square roots of perfect square numbers, where multiple solutions may exist. For example, students can: Justify solutions convincingly. Eg use digits 2, 0, 1, 9 to develop number sentences which are equivalent to 4 incorporating at least one square or square root, such as: $1^2 + 0 + \sqrt{9} = 4$	Students systematically and efficiently solve problems involving squares of numbers and square roots of perfect square numbers above 20^2. For example, students can: Justify solutions through multiple representations, highlighting connections and efficiencies. Eg using your understanding of square numbers determine 23^2, providing multiple representations to support your answer, without using a calculator (possibilities may include using a region multiplication model; starting from 20^2, and repeatedly adding $(n + (n + 1))$ to find the next square and continuing the process.

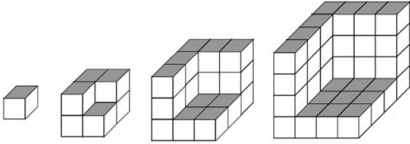
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
				If $50^2 = 2\,500$, then find 49^2 without using a calculator
	With support, students solve problems involving addition and subtraction of integers. For example, students can: Explain why one integer is larger than another eg Place -2 and -5 on a number line, and find the difference between them.	Students solve problems involving addition and subtraction of integers. For example, students can: - Explain the difference between two integers. - Student can explain why addition and subtraction produces specific results eg Calculate the following, showing your answers on a number line, or other materials such as two-coloured counters: a) $-4 + 5$ b) $5 + -2$ c) $-1 - -7$ Eg the temperature in Launceston is currently -4°C. If the temperature warms by 7°C, then what will the temperature be?	Students solve multi-step or worded problems involving addition and subtraction of integers. For example, students can: Work forwards and backwards from given elements and can clearly explain their thinking. Eg I begin with a number, I add 9, then subtract 5. The answer I get is 1. What number did I start with? Show how you reached your answer.	Students systematically solve problems involving addition and subtraction of integers, where multiple solutions may exist. For example, students can: Communicate and justify a systematic collection of solutions. Eg using the integers from -9 to 9 at most one time each, fill in the boxes to create all possible equations. OR Where each side has the greatest/least possible value. $-\square + \square = \square - (\square)$
	Students use some all 4 operations in calculations involving positive fractions and decimals to tenths, choosing efficient calculation strategies. For example, students can: - Calculate straightforward addition and subtraction of fractions, eg Jeffrey has $\frac{1}{8}$ of a pizza and Jemima has $\frac{2}{8}$ of a pizza. How much do they have altogether? - Calculate or make a model to determine 0.3×0.2	Students use all 4 operations in calculations involving positive fractions and decimals, choosing efficient calculation strategies. For example, students can: - Use a range of strategies and representations to solve problems, and explain which of the strategies if most efficient, when working with fractions and decimals, using appropriate rounding and estimation to check the reasonableness of solutions. - Use common denominators to solve a range of fraction problems,	Students use all 4 operations in calculations involving positive fractions and decimals, including mixed numbers and representations, choosing efficient calculation strategies, including finding original quantities when a fractional amount is known. For example, students can: Efficiently calculate, fluently moving between different or mixed representations of fractions and decimals, and simplifying solutions. Eg calculate:	Students use all 4 operations in multi-step calculations involving positive fractions and decimals, including mixed numbers and representations, and correctly using order of operations, choosing and justifying efficient calculation strategies. For example, students can: Communicate connections in their systematic approach and apply sound reasoning to justify solutions, such as converting all rational numbers into equivalent representations.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard								
		with related and unrelated denominators, including division, and simplifying solutions. Eg Joni has $\frac{3}{4}$ of a pizza left over for lunch. She wants to share it equally with her friend. Determine how much of the pizza each person should get. $\frac{1}{2} \text{ of } \frac{6}{8} = \frac{3}{8}$ $\frac{3}{4} = \frac{6}{8}$ $\frac{6}{8} - \frac{3}{8} = \frac{3}{8}$ 3 eighths plus 3 eighths is equivalent to 3 fourths. 6 eighths divided by 2 is 3 eighths. $\frac{1}{2} \times \frac{3}{4} = \frac{1 \times 3}{2 \times 4} = \frac{3}{8}$	$0.6 \times 2\frac{3}{8}, \frac{3}{5} \times 5.3$ $2.6 + \frac{2}{3}, \frac{3}{5} + 0.6$ $3\frac{2}{3} - 0.3, 5.6 - \frac{4}{5}$ $\frac{3}{5} \div \frac{3}{10}, 5.6 \div 0.3$ 2 fifths of a full bag of dog food remains in the bag and currently weighs 7.6 kg. How much did the bag weigh when full? - Explain how you carried out your calculation and why you chose this strategy. - Use and justify more than one strategy to solve division of a positive rational number by a fractional amount, for example using a number line, region model, or written strategy.	eg $0.8 - \frac{1}{4} \times 2\frac{2}{3}$ $\left(0.8 - \frac{1}{4}\right) \div 2/3$ Finding common denominator of all three or more fractions before calculating, eg $\frac{1}{8} + \frac{11}{12} - \frac{1}{3}$								
	With support, students choose between equivalent representations of rational numbers and percentages to assist in calculations. For example, students can: Complete the following table, and use these values in calculations: <table><tr><th>Fraction</th><th>Decimal</th><th>Percentage</th></tr><tr><td>$\frac{1}{2}$</td><td></td><td></td></tr><tr><td></td><td>0.1</td><td></td></tr></table>	Fraction	Decimal	Percentage	$\frac{1}{2}$				0.1		Students choose between equivalent representations of rational numbers and percentages to assist in calculations. For example, students can: - Demonstrate more than one approach to solving, explains preference. Eg to calculate 12.5% of 40, you could choose to do any of the following: $12.5\% \times 40$ 0.125×40 $\frac{1}{8} \times 40$	Students choose between equivalent representations of rational numbers and percentages to assist in calculations, justifying their choices and strategies. For example, students can: Recognise and explain advantages of calculation and number properties (eg commutativity and associative laws and equivalent representations) in describing their solution. Eg how many different ways could you calculate $\frac{2}{5}$ of $\frac{1}{2}$? Which way is most efficient? Why?
Fraction	Decimal	Percentage										
$\frac{1}{2}$												
	0.1											

Strand	Approaching the Standard			At Standard	Above Standard	Well Above Standard
			20%	Which would you choose? Explain your choice. Student might choose $\frac{1}{8} \times 40$ and explain that it is easier to divide 40 by 8 to get the answer than completing the other operations.	a) $\frac{1}{2}$ of $\frac{2}{5} = \frac{1}{5}$ b) 50% of $\frac{2}{5} = \frac{1}{5}$ c) $\frac{1}{2}$ of $\frac{4}{10} = 0.2$	Is the method you chose the most efficient? How do you know? Eg state equivalent fractions as tenths, convert to decimals. a) $(0.6 + 1.5 - 0.9)$ • Explain efficiencies over converting to common denominators by reducing the additional step in converting the mixed number to an improper fraction.
	Students use provided mathematical modelling to solve practical problems involving rational numbers, percentages and ratios, in financial and other applied contexts, partially justifying choices of representation. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: a) translating the problem into maths b) using maths to solve the problem c) interpreting and communicating the solution back in the real-world context d) justification of the mathematical representations. - Describe the process that they are following when using a given model, and may describe a limitation of the given representation.			Students use mathematical modelling to solve practical problems involving rational numbers, percentages and ratios, in financial and other applied contexts, justifying choices of representation. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: a) translating the problem into maths b) using maths to solve the problem c) interpreting and communicating the solution back in the real-world context d) justification of the mathematical representations. - Identify possible models to solve and justify their chosen representation. - Determine the percentage profit/loss of a class fundraising activity sausage sizzle.	Students use mathematical modelling to solve practical problems involving rational numbers, percentages and ratios, in financial and other applied contexts, justifying choices of representation, and evaluation of solutions. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: a) translating the problem into maths b) using maths to solve the problem c) interpreting and communicating the solution back in the real-world context d) justification of the mathematical representations e) evaluation with some valid reasons and checking of answers. - Compare the percentage profit/loss of a class fundraising sausage sizzle, if the sausages were sold for a different price, ie at the event sausages were sold for \$1.50, what would the percentage profit/loss have been if	Students use mathematical modelling to solve practical problems involving rational numbers, percentages and ratios, in financial and other applied contexts, justifying choices of representation, with reasoned evaluation of solutions. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: a) translating the problem into maths b) using maths to solve the problem c) Interpreting and communicating the solution back in the real-world context d) justification of the mathematical representations e) reasoned evaluation and confirmation of answers. - From a targeted percentage profit of a class fundraising activity sausage sizzle, determine the

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	Eg students are given a graph displaying the quantity of red paint required to produce pink paint at a ratio of 1:4, red to white and identify that the model as presented is limited by the range of the independent variable.		the sausages were sold instead for \$2.	selling price for a quantity of sausages to meet that target.

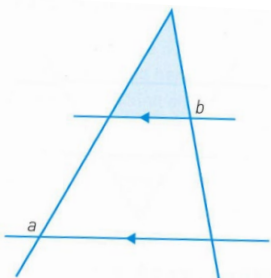
Algebra

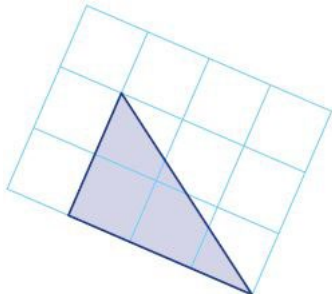
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Algebra	Students use given algebraic expressions to represent situations. For example, students can: - Calculate solutions when using given expressions that represent familiar situations. Eg using the area formula/deriving a or applying perimeter of a rectangle $A = w \times l$ $P = l + l + w + w$ - or $P = 2l + 2w$	Students use algebraic expressions to represent situations. For example, students can: - Calculate solutions choosing appropriate formula. - Formulate algebraic formulas to represent familiar situations. - Use variables to link attributes and measures being modelled. Eg scoring in AFL. $p = 6g + b$ to describe a total of points expressed as goals (worth 6 points) and behinds (worth one point).	Students use and manipulate algebraic expressions to represent situations. For example, students can: - Use and create algebraic expressions to represent complex patterns and solve, eg matchsticks or dot patterns. - Create equivalent algebraic expressions that represent the same situation.	Students use and manipulate algebraic expressions to represent complex situations. For example, students can: Create rules independently and manipulate algebraic expressions to represent complex patterns, that may include non-linear components, eg growing patterns that includes increasing squares. 
	Students describe some relationships between variables from authentic data. For example, students can: Describe some relationships (as the variable is substituted with increasing numbers, the total cost goes up). Eg $Cost = \\$40 \text{ per widget}$	Students describe the relationships between variables from authentic data. For example, students can: - Explain how the relationship between variables can influence data. For example, using an hourly rate (r) may impact cost. Eg use a flat hourly rate and call out fee of a plumber; \$80/hr plus a \$50 call out fee. Students describe how the cost will change depending on how long the plumber works. - Interpret graphs of authentic data travel time, power usage.	Students describe the relationships between variables from authentic data and make reasoned predictions beyond the range of the displayed data. For example, students can: Make predictions beyond the displayed data range through the use of models that describe relationships between authentic data (climate data, human population, vaccination rates).	

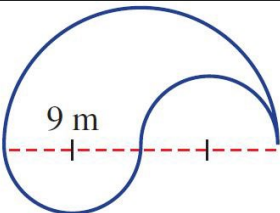
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Algebra (c)	Students substitute values into simple formulas, without co-efficients, to determine unknown values. For example, students can: Substitute values into simple formulas, with no coefficients, to determine unknown values, such as: $\text{Cost} = x + 17$	Students substitute values into formulas to determine unknown values. For example, students can: - Use the formulas to find area or volumes of 2D shapes and 3D objects with or without digital technologies. - Use digital technology to examine the effect of changing variables, such as modifying a side length and examining the effect upon the area/volume. - Create algebraic expressions to describe operations from “function machines” or visual patterns using variables, constants and the effect of brackets.	Students manipulate formulas and substitute values to determine unknown values. For example, students can: Manipulate formulas to find unknown values, that require more than 2 steps, eg determine a range of heights and widths for a rectangular prism, given a volume and length.	
	With support, solves linear equations with natural number solutions. For example, students can: Solve linear equations that will have natural number solutions (1, 2, 3, ...). Eg solve the following equation, using the balancing provided: $\begin{array}{rcl} x + 6 & = & 25 \\ -6 & & -6 \end{array}$	Solve linear equations with natural number solutions. For example, students can: Using strategies including concrete materials, the balance model and backtracking to solve linear equations, describing the process, and verifying solutions by substitution. eg $2x - 5 = 17$.	Solve linear equations with natural number solutions that involve more than 2 steps. For example, students can: Solve linear equation that involve more than 2 steps, so may have the same variable on both sides of the equation, and verifying solutions by substitution eg: $2x + 17 = 5x - 4$ $\frac{3x + 2}{5} = 7$	Solve linear equations with natural number solutions that involve more than 2 steps, and have rational terms or coefficients. For example, students can: Manipulate formulas to find unknown values, including solutions requiring 2 or more steps, so may have the same variable on both sides of the equation and have rational terms or coefficients, verifying solutions by substitution eg: solve the following equations: $\frac{5x}{2} - 4 = x + 8$ $\frac{3x + 2}{5} = 7$

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Algebra	With support, create tables of values related to algebraic expressions and formulas, and describe the effect of variation of a single variable. For example, students can: - Complete the following table for the rule: $n = 9 \times a$: <table border="1"><tr><td>a</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>n</td><td></td><td></td><td></td><td></td><td></td></tr></table> - Plot this on the Cartesian plane, describe the shape of the graph and the effect of changing the value of a.	a	0	1	2	3	4	n						Students create tables of values related to algebraic expressions and formulas, and describe the effect of variation. For example, students can: - Create tables of values for linear functions (such as function machines, visually growing patterns, the circumference of a circle), plot them on the Cartesian plane and describe the shape of the graph. - Recognise that linear functions may be represented in a variety of ways, including as a table of values, a rule or formula, a written or verbal description and a graph. - Use digital tools, such as Excel, to manipulate formulas with more than one variable, describing the effect of variation of a variable. Eg the volume of a rectangular jewellery box is given by the formula $V = l \times w \times h$. The length and width of the box are both 10 cm. Student explains, for example “the volume doubles if the height doubles, because the volume is multiplied by same factor as the height.”	Students create tables of values related to algebraic expressions and formulas from a range of representations, and describe and explain the effect of variation. For example, students can: - Recognise that linear functions may be represented in 4 ways: as a table of values; a rule or formula; word form; and a graph; and create a table of values from any of the other 3 representations - Use digital tools, such as Excel, to manipulate formulas with more than one variable, describing the effect of variation of 2 variables. Eg the volume of a rectangular jewellery box is given by the formula $V = l \times w \times h$. The length of the box is 10 cm. Student explains, for example “the volume increases by a factor of 4 if the height and width both double, because the volume is multiplied by 2, two times.”	Students create tables of values related to algebraic expressions and formulas, from a range of representations and describe and generalise the effect of variation. For example, students can: - Recognise that linear functions may be represented in 4 ways: as a table of values; a rule or formula; word form; and a graph; and from a table of values create valid representations other than a graph. - Use digital tools, such as Excel, to manipulate formulas with more than one variable, describing the effect of variation of 2 variables. Eg the volume of a rectangular jewellery box is given by the formula $V = l \times w \times h$. The length of the box is 10 cm. Student generalises, for example “multiplying any dimension by an amount changes the volume by that multiple. Multiplying more than one dimension by an amount, multiplies the effects together, (giving examples)”.
a	0	1	2	3	4											
n																

Measurement

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard												
Measurement	With support, students apply knowledge of angle relationships and the sum of angles in a triangle to solve problems, giving reasons. For example, students can: - Determine the sum of internal angles in a triangle relates to familiar quadrilaterals using concrete materials. - Complete a table relating the number of sides of polygons to the sum of the interior angles.	Students apply knowledge of angle relationships and the sum of angles in a triangle to solve problems, giving reasons. For example, students can: - Apply knowledge of sum of interior angles in a triangle and angles formed when a transversal crosses parallel lines to solve unknown angle problems in familiar 2D shapes and contexts eg: - Exploring and generalising the angles in a quilting pattern. - Using the angle sum of a triangle to determine the interior angle sum of quadrilaterals, pentagons and hexagons, and can describe the effect upon the interior angle sum of increasing the number of sides. - Demonstrate and describe how angle properties are used in the design and use of car jacks, scissor lifts or similar equipment.	Students transfer knowledge of angle relationships and the sum of angles in a triangle to solve problems, giving reasons. For example, students can: - Clearly explain why the exterior angle of a triangle is equal to the sum of the non-adjacent interior angles and solve related problems. - Generalise a rule to determine the interior angle sum in an n-sided polygon, and solve related problems. - Determine and justify several sets of answers to more complex questions such as:	Students transfer knowledge of angle relationships and the sum of angles in a triangle to solve complex problems, giving reasons. For example, students can: - Geometrically reason connections between internal and external angles in multiple complex situations. - Use sound geometric reasoning to solve problems that combine triangles and parallel lines, such as any of the problems on page 94 of Sullivan, P. (2001). Building Engagement in Middle Years Mathematics, Find the missing angles eg:												
	<table><tr><th>Polygon</th><th>Sum of interior angles</th></tr><tr><td>Triangle</td><td>180°</td></tr><tr><td>Parallelogram</td><td></td></tr><tr><td>Rectangle</td><td></td></tr><tr><td>Regular pentagon</td><td></td></tr><tr><td>Irregular pentagon</td><td></td></tr><tr><td>Regular hexagon</td><td></td></tr></table>	Polygon	Sum of interior angles	Triangle	180°	Parallelogram		Rectangle		Regular pentagon		Irregular pentagon		Regular hexagon		 Angles <i>a</i> and <i>b</i> are both multiples of 5 and add up to 225 degrees. Work out possible values for each of the angles in the shaded triangle. Give at least two answers. Sullivan, P. (2001). Building Engagement in Middle Years
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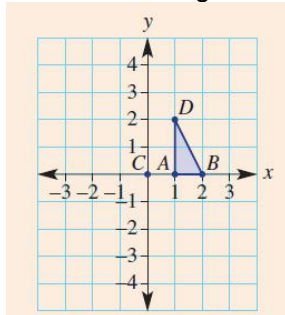
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	With support, students use formulas for the areas of triangles and parallelograms and the volumes of rectangular and triangular prisms to solve problems. For example, students can: - With concrete materials, including gridded paper, find the area of right-angle triangles, and make some connections to the area of the surrounding rectangle eg: - Find the area of this triangle.  - Create rectangular prisms from unit blocks to find the volume.	Students use formulas for the areas of triangles and parallelograms and the volumes of rectangular and triangular prisms to solve problems. For example, students can: - Given their perpendicular heights of any triangle, establish that the area of a triangle is half the area of an appropriate rectangle by using the spatial relationship between rectangles and different types of triangles, and solve related problems. - Solve simple problems related to the volume of rectangular and triangular prisms.	Mathematics, Find the missing angles p. 93 Students use and manipulate formulas for the areas of triangles and parallelograms and the volumes of rectangular and triangular prisms to solve problems. For example, students can: - Find unknown dimensions, given the: - Area and one dimension of a triangle and parallelogram. - Volume and 2 dimensions of rectangular and triangular prisms. - Generalise the formula for the area of right prisms to find the volume of other right prisms.	the size of angle g? Give at least two answers Students use and manipulate formulas for the areas of triangles and parallelograms and the volumes of rectangular and triangular prisms to solve complex problems that may have more than one solution. For example, students can: Find a range of possible solutions given the volume of a rectangular or triangular prism and one dimension and other right prisms where appropriate.
	With support, students describe the relationships between the radius, diameter and circumference of a circle. For example, students can: Locate and label the radius, diameter and circumference of a circle.	Students describe the relationships between the radius, diameter and circumference of a circle. For example, students can: Label the parts of a circle including centre, radius, diameter, circumference and use one of radius, diameter or circumference	Students describe the relationships between the radius, diameter and circumference of a circle, and solve problems involving unit fractional parts of the circumference. For example, students can: Find the perimeter of this shape using knowledge of circumference of circles.	

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	- Identify that the diameter of a circle is twice the radius. - Follow a procedure to calculate the circumference of a circle given the radius or diameter.	to determine the measure of the other 2. Recognise π as the constant in the proportional relationship between the circumference of a circle and its diameter.		

Space

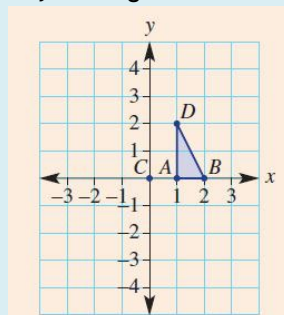
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Space	With support, students classify polygons according to their features and create a simple algorithm designed to sort and classify shapes. For example, students can: - Classify polygons according to a stated property such as the presence of parallel lines, equal angles or equal side length. - Create and apply a simple algorithm to sort shapes, such as the number of angles or the number of sides.	Students classify polygons according to their features and create an algorithm designed to sort and classify shapes. For example, students can: - Describe relationships between different quadrilaterals. - Given a range of shapes, sort into categories defined by the student, drawing on a range of features, using flowcharts or hierarchies.	Students classify polygons according to their features and create and review an algorithm designed to sort and classify shapes. For example, students can: Argue and justify that a polygon may be classified in more than one way, or belong in more than one category. Eg a square is a rectangle because ..., though not all rectangles are squares because ...	Students classify polygons according to their features and create, review and refine an algorithm designed to sort and classify shapes. For example, students can: Refine an algorithm so that they can comprehensively justify why for example, a square is also a rectangle, rhombus, kite, parallelogram, and trapezium.
	With support, students represent objects two-dimensionally in different ways, describing the usefulness of these representations. For example, students can: Construct a 3D object from a net and create views of objects (options include front, back, side, top and bottom), and can describe where these representations may be useful.	Students represent objects two-dimensionally in different ways, describing the usefulness of these representations. For example, students can: - Determine which nets will make a cube, rectangular prism, triangular prism or pyramid. - Build objects by interpreting isometric and perspective drawings. - Use isometric and square grid paper to draw views of front, back, side, top and bottom of objects.	Students represent objects two-dimensionally in different ways, describing and justifying the usefulness of these representations. For example, students can: - Convincingly justify that a particular two dimensional representation is more useful or appropriate than others in particular circumstances, for example, why isometric views may be preferred to perspective drawings. - Create non-examples of nets and justify why they are not valid.	Students represent objects two-dimensionally in different ways to solve problems describing and justifying the usefulness of these representations. For example, students can: - Recreate 'missing' views and representations from a partial set and justify why the created views are valid. - Generalise how objects can be represented in two dimensions and describing how representations.
	Students use coordinates to describe a simple (eg translation) transformations of points in the plane. For example, students can: Give the coordinates of a simple shape on the Cartesian plane, and	Students use coordinates to describe transformations of points in the plane. For example, students can:	Students use coordinates to describe multiple transformations of points in the plane. For example, students can:	

then the revised coordinates after a small translation eg:



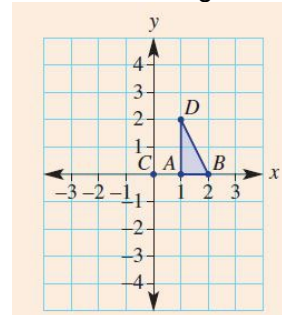
- Draw the image of this shape and give coordinates of A', B' and D' after it is translated two steps upwards. Describe how the shape has changed.

- Correctly describe coordinates after a single transformation other than a translation parallel to either the x or y axes eg:



- Draw the image of this shape and give coordinates of A', B' and D' after it is rotated 90° anticlockwise about C. Describe how the shape has changed.

- Correctly describe coordinates after 2 transformations eg:



- Draw the image of this shape and give coordinates of A', B' and D' after it is rotated 90° anticlockwise about C, and reflected about the x -axis. Describe how the shape has changed.

Statistics

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, students plan and conduct statistical investigations involving discrete and continuous numerical data, using appropriate displays. For example, students can: - Undertake an investigation using primary or secondary data, including relevant investigations in HASS or Science where data may be collected in a survey or experiment. - Create data displays. - Report some findings in terms of the shape and summary statistics.	Students plan and conduct statistical investigations involving discrete and continuous numerical data, using appropriate displays. For example, students can: - Plan and undertake an investigation using primary or secondary data, including relevant investigations in HASS or Science where data may be collected in a survey or experiment. - Analyse and interpret distributions of data. - Report findings in terms of the shape and summary statistics.	Students plan and conduct statistical investigations involving discrete and continuous numerical data, using appropriate displays, justifying which display best represents the data. For example, students can: - Plan and undertake an investigation using primary or secondary data, including relevant investigations in HASS or Science where data may be collected in a survey or experiment. - Analyse and interpret distributions of data, justifying which data display best represents the data. - Report findings in terms of the shape and summary statistics.	Students plan and conduct statistical investigations involving discrete and continuous numerical data, using appropriate displays, justifying which display best represents the data, and making inferences. For example, students can: - Plan and undertake an investigation using primary or secondary data, including relevant investigations in HASS or Science where data may be collected in a survey or experiment. - Analyse and interpret distributions of data, justifying which data display best represents the data. - Report findings in terms of the shape and summary statistics, making inferences.
	With support, students interpret data in terms of the shape of distribution and summary statistics, identifying possible outliers. For example, students can: - Interpret data in terms of the shape of distribution. - Determine a measure of central tendency. Eg could use a stem and leaf plot to help connect features of the data display and can discuss the location of the mode and/or median of these distributions.	Students interpret data in terms of the shape of distribution and summary statistics, identifying possible outliers. For example, students can: - Connect features of the data display, such as, highest frequency, clusters, gaps, symmetry or skewness, to the mode, range and median. - Describe the shape of distributions using terms such as positive skew, negative skew, symmetric and bi-modal and discuss the location of the median,	Students interpret and compare data sets in terms of the shape of distribution and summary statistics, identifying possible outliers. For example, students can: Make comparisons between data displays for different data sets, and can justify differences using appropriate references to the features, shape and summary statistics.	Students interpret and compare data sets in terms of the shape of distribution and summary statistics, identifying and describing the effect of possible outliers. For example, students can: Make comparisons between data displays for different data sets, and can justify differences using appropriate references to the features, shape and summary statistics and explain how outliers impact data summaries.

		mode and mean for these distributions. Compare the mean, median and mode of data sets with and without outliers		
	With support, students decide which measure of central tendency is most suitable and explain their reasoning. For example, students can: Define and calculate a measure of central tendency of a numerical data set.	Students decide which measure of central tendency is most suitable and explain their reasoning. For example, students can: - Calculate range, median, mode and mean for discrete and continuous data sets. - Decide which measures of central tendency are useful and explain their reasoning. - Experiment with data sets by adding or otherwise modifying data values, and describing the impact upon measures of central tendency.	Students flexibly work with measures of central tendency to construct valid datasets and explain their reasoning. For example, students can: Create one or more data sets that could match a data summary. Eg in the first five games of the season the school's basketball team's scores had a median of 36 points, and a mean of 40 points. What might be the scores in each of the five games?	Students flexibly and creatively work with measures of central tendency to construct valid datasets and explain their reasoning. For example, students can: - Create one or more data sets that could match a data summary, and modify the data set as the summary changes. Eg in the first five games of the season the school's basketball team's scores had a median of 36 points, and a mean of 40 points. What might have the scores in each of the five games? - After five more games the mean rose to 43 points and the median was unchanged. In the sixth game they scored a season-high 85 points. What might be the other scores for the second set of five games?

Probability

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard												
Probability	With support, students list sample spaces for single step experiments, using language of chance assign probabilities to outcomes and predict relative frequencies for related events. For example, students can: - List sample spaces for single step experiments, use words to describe possible outcomes. - Eg spinning a spinner split into quarters, with 2 green sectors 1 red sector and 1 blue sector, using the language of probability to describe outcomes.	Students list sample spaces for single step experiments, assign probabilities to outcomes and predict relative frequencies for related events. For example, students can: - Correctly list samples spaces and assign probabilities for single step experiments, such as tossing a coin, throwing a dice, a spinner divided into equal sized parts. - Recognises that the chance of something occurring or its complement has a total probability of one (eg the probability of rolling a 3 is $\frac{1}{6}$ and the probability of not rolling a 3 is $\frac{5}{6}$). - Use assigned probabilities to predict relative frequencies for related events (related events are multiple trials of single step experiments). Eg assigning the probability for throwing a 6 on a dice and using this to predict the number of times a 6 will occur when a dice is thrown 100 times. <table><tr><th>Roll</th><th>Probability</th><th>Predicted frequency</th><th>Predicted relative frequency</th></tr><tr><td>1</td><td>1/6</td><td>20</td><td>0.2 or 20%</td></tr><tr><td>2</td><td>1/6</td><td>11</td><td>0.11 or 11%</td></tr></table>	Roll	Probability	Predicted frequency	Predicted relative frequency	1	1/6	20	0.2 or 20%	2	1/6	11	0.11 or 11%	Students list sample spaces for single step experiments that they create, assign probabilities to outcomes and predict relative frequencies for related events. For example, students can: - Correctly list samples spaces and assign probabilities for single step experiments that they create, such as a spinner with 20% yellow, 45% blue and 35% green. - Use assigned probabilities to predict relative frequencies for related events for their spinner (related events are multiple trials of single step experiments). - Clearly explain why tossing a standard, non-biased coin may result in runs of 5 or more tails (or heads) in a row.	
Roll	Probability	Predicted frequency	Predicted relative frequency													
1	1/6	20	0.2 or 20%													
2	1/6	11	0.11 or 11%													

		3	1/6	22	0.22 or 22%		
		4	1/6	13	0.13 or 13%		
		5	1/6	15	0.15 or 15%		
		6	1/6	19	0.19 or 19%		
		Total		100			
	With support, students conduct repeated single-step chance experiments and run simulations using digital tools, giving some reasons for differences between predicted and observed results. For example, students can: Flip a coin twenty times and record the results in a table and then run a simulation for 100 flips, and identify differences between the two data sets.	Students conduct repeated single-step chance experiments and run simulations using digital tools, giving reasons for differences between predicted and observed results. For example, students can: - Roll a 6-sided dice 60 times, then use an online simulator (such as Dice Roll Simulation – GeoGebra) and compare the results for 600, 6 000 and 60 000 rolls. - Describe the difference being related to the number of trials, that is, that theoretical probabilities for seemingly random events will only be approached if the number of trials is large enough.				Students design and conduct repeated single-step chance experiments and run simulations using digital tools, giving detailed reasons for differences between predicted and observed results. For example, students can: - Create a spinner with 20% yellow, 45% blue and 35% green, run a number of trials and then a much large number in a simulation, and clearly explain the differences between the trial and simulations. - Explain, for instance, that a small number of outcomes will not necessarily be soon offset by a matching set (the gambler’s fallacy).	